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RESEARCH ARTICLE

Revealing youth-perceived cultural ecosystem services for high-density urban green space management: a deep learning spatial analysis of social media photographs from central Beijing

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Abstract
Context In high-density urban areas, managing urban green spaces (UGSs) is challenging due to limited space. Young people in these environments often experience mental health issues, exacerbated by insufficient exposure to green spaces. UGSs provide cultural ecosystem services (CESs), crucial for enhancing mental well-being and landscape sustainability in general.
Objectives Focusing on central Beijing, our study addresses three key questions: (1) What is the spatial pattern of youth-perceived CESs? (2) Which CESs are most valued by youth? and (3) How do the various UGSs provided by different landscape features contribute to youth-perceived CESs ?
Methods We used geo-tagged images from Little Red Book, the most active social platform for young people, combined with deep learning techniques to map the spatial distribution of nine types of CES perceived by youth. Random forest analysis was used to identify the contribution of different landscape features to these CESs.
Results Our findings show that social recreation is the most valued CES, followed by nature appreciation. Social and artistic CESs are more clustered, while nature appreciation is more dispersed. UGSs in built environments play a more significant role in supporting social recreation, while urban parks contribute primarily to landscape and nature appreciation. Waterbodies and parks contribute less to social recreation.

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Conclusions This study highlights how crowd-sourced imagery and deep learning methods can reveal the spatial distribution of CESs and their potential to improve youth mental health. The findings offer valuable insights for urban planners to enhance CESs in UGSs, addressing mental health concerns in high-density urban settings.

Keywords Cultural ecosystem services · Social media photographs · Deep learning · Urban green spaces · Youth

Introduction

Urban green spaces (UGS) play a crucial role in high-density urban areas, offering residents a connection to nature (Wolch et al. 2014). In high-density megacities, UGS serve as critical infrastructure for delivering cultural ecosystem services (CESs) that promoting human health and well-being (Bolund and Hunhammer 1999; Elmqvist et al. 2015; Tzoulas et al. 2007). CES encompasses non-material benefits derived from ecosystems, including aesthetic qualities, cultural heritage, recreational opportunities, and social interactions, which are critical for mental health outcomes (Christine 2015; Milcu, Hanspach et al. 2013; Xu et al. 2020; Plieninger et al. 2022). These CES are directly derived from people's perceptions and experiences of landscapes. As global urbanization accelerates, assessing the diverse CES provided by UGS has become imperative for developing targeted quality improvement strategies. This is particularly critical in high-density megacity contexts, where population growth pressures and land scarcity-driven land-use competition threaten the effectiveness of UGS planning and management.

Assessing CES presents challenges due to their non-material nature (Daniel et al. 2012; Kirchhoff 2012). Early methods included monetary valuation techniques and socio-cultural preference assessments through interviews (Klain et al. 2014), surveys (Iniesta-Arandia et al. 2014, Oteros-Rozas et al. 2014), focus groups (Norton et al. 2012), Visitor Employed Photography (VEP) and Public Participatory mapping (PPGIS) (Plieninger et al. 2013). In recent years, utilizing crowdsourced geospatial data from social media platforms like Flickr, Panoramio, and Instagram has for evaluating CES has attracted increasing attention due to its widespread availability, frequent updates and large sample sizes compared with traditional methods (Xu et al. 2023; Casalegno et al. 2013; Gliozzo et al. 2016; Richards and Friess 2015; Tenerelli et al. 2016; van Zanten et al. 2016;). The users' generated contents posted on social media platform are widely regarded as important proxies for people's perceptions, offering advantages such as reflecting large public preferences and providing precise service provision information at specific locations (Czembrowski et al. 2016; Bagstad et al. 2014). Thus, in the context of high-density urban areas, assessing public perception of CES through social media-based content analysis demonstrate unique advantages in spatial–temporal scale coverage and sample size. Recent studies have validated the effectiveness of social media photographs in capturing cultural ecosystem services (CES) perceptions (Olafsson et al. 2022; Cardoso et al. 2022; Ghermandi et al. 2022. For instance, Olafsson et al. (2022) compared CES results from Flickr photo analysis and visitor surveys across 19 cities, reporting high consistency in the assessment of aesthetic and recreational values. Similarly, Cardoso et al. (2022) demonstrated that deep learning models could effectively classify CES-relevant content from social media images, further confirming the potential of social media as a reliable tool for understanding CES. Ghermandi et al. (2022) also reported 87.5% agreement (Cohen's kappa = 0.54) between CES studies based on Flickr photo content classification and those based on text. Through these comparisons and validations, these studies have shown that social media photograph-based analysis is a reliable approach for capturing spatially explicit CES perceptions, particularly in high-density urban areas.

Young people (aged 18–34) in these high-density megacity contexts face unique challenges for mental health, including chronic stress, social isolation, and sensory overload. Globally, WHO demonstrates that urban young people face high mental health burdens, with restorative environment deprivation as key factor contributing to these cases. Numerous studies confirm the therapeutic potential of UGSs—through mechanisms such as attention restoration, social cohesion enhancement, and stress mitigation (Bratman et al. 2019)—the effectiveness of such interventions depends critically on users' ability to access and derive specific CES. mechanisms ranging from attention restoration to social cohesion enhancement. Alarmingly, urban young people generally have insufficient utilization of and exposure to green spaces. Globally, youth in high-density cities average only 18 min/day of quality green exposure, 63% below the WHO-recommended 50 min for mental health (WHO 2023; Wang et al. 2022; Xu et al. 2024). This data reveals a critical paradox: increasing green space accessibility alone cannot resolve the mental health crisis in dense urban areas. To bridge this gap, there is an urgent need to assess the youth perceived CESs provided by the UGSs in high-density megacities to inform spatially optimized UGS planning strategies that address both physical and psychological well-being outcomes.

However, empirical studies focusing on youth-perceived CESs delivered by UGSs remain scarce. Current CES research predominantly relies on age-integrative assumptions, overlooking the fundamental differences in service access, perception, and value attribution among age groups (Plieninger et al. 2022). Even though scholars noticed that individual modes of environmental experiences, behaviors and preferences significantly influence the CES they perceive, more studies focused on the behaviors and preferences of different age groups and there are little existing studies had addressed that youth perception CES differ from the older group in the parks (e.g. gained more CESs related to social interactions and exercises) (Zhang et al. 2022). But these little existing studies about youth perceived CES were based on the traditional approaches like surveys and VEP are limited by small sample sizes and failure to capture youth's unique experiences to nature in the social media era (Edwards and Larson 2020). Pervious studies addressed that the urban youth perceptions and experiences of nature were mediated by the digital media such as social media and phone (Edwards and Larson 2020). In megacities, young people engage with UGSs through social media users' generated content, such as sharing moments of serenity, documenting scenic vistas, or recounting joyful memories in parks), for self-expression, dentity construction and embedding nature into their value systems (Arts et al. 2021). Youth's CES perceptions are increasingly shaped by digital narratives—not just physical visits but virtual engagements through photo-sharing and hashtag usage (Arts et al. 2021; Duckett et al. 2021). This shift has critical implications for UGS planning in megacities, where youth-oriented UGSs management must account for digitally mediated CES values. Therefore, it is demanding to combine with social media approaches for assessing youth perceived CES in high-density urban contexts for supporting UGSs planning and management.

Under these contexts, the aim of this article is to explore the youth-perceived CESs from diverse UGSs in a high-density urban context via coming social media approaches. We used the central Beijing, one of the highest-density megacities in Asia as study area. In China, young people have higher average levels of anxiety than other adult age groups, and this situation is even more pronounced in megacities, like Beijing (Fu et al. 2023). In central Beijing, urban renewal initiatives provide a critical opportunity to align UGS quality improvements with youth-centric CES values, addressing both physical and psychological well-being. In this study, we assessed the youth-perceived CESs based on the crowdsourced data from the most active social media platforms among Chinese young people. Combing with deep learning techniques and GIS spatial analysis, we identified hotspots of youth-perceived CESs as well as the spatial correlation between their perceived CES for supporting the optimization of UGSs in central Beijing as a high-density urban context. It has reference significance for the UGSs management in the high density urban context of other countries.

To fulfill the aim of this study, we put forward three research questions:

1. Which type of CES is perceived the most by the general youth?
2. What's the spatial pattern of the CES that youth perceived in central Beijing?
3. What's is the respective contribution of various UGSs in central Beijing for youth perceived each CES?

Method and data

Case study

Our research focuses on central Beijing's UGS (Fig. 1A, B). Beijing, China's greatest high-density megacity, has grown rapidly in the past two

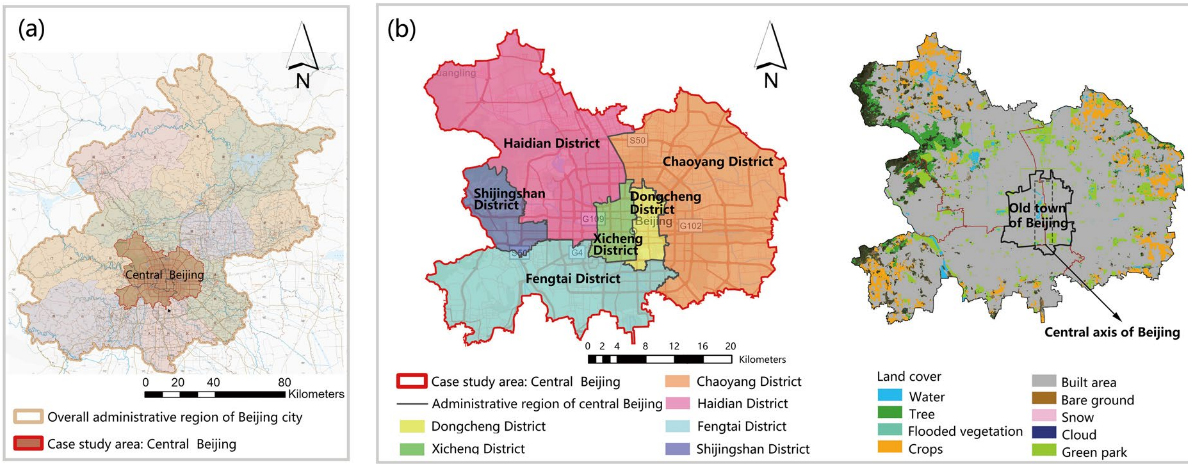


Fig. 1 Location of the overall administrative region of Beijing, central Beijing (comprising six districts) in China and the land cover features

decades. This population growth presents concerns for Beijing's urban green space management (Wang et al. 2023). Central urban areas have the most challenging issues. The central urban area has six districts: Dongcheng, Xicheng, Chaoyang, Haidian, Fengtai, and Shijingshan (Fig. 1C). Considering Beijing's central metropolitan area's high population density, green space efficiency is crucial for public resource allocation and use. Despite the fact that the current per capita park green space in central Beijing is 16.7 m^2 (Capital Construction Network 2020), the dearth of fine-grained management of green space prevents the public from effectively utilizing all urban green space (Liu et al. 2021). Problems such as substandard green space quality and difficulty in public perception of CES continue to be pervasive and have a significant impact on the efficiency of green space utilization.

Research process

The study is divided into four major sections.

Initially, social media data was collected by crawling the Little Red Book (LRB) the largest Chinese life-sharing social media platform, through Python and keywords. Based on the notes that people uploaded on LRB about their perception of nature in central Beijing, we screened and collected the geo-tagged photographs to establish the public perception of the UGSs dataset.

Secondly, the collected geo-tagged photographs were classified by the deep learning model according to various CES types, and therefore each geo-tagged photograph was assigned to one CES category.

After that, the kernel density analysis was utilized to investigate the CES spatial distribution.

Last but not least, to clarify the contribution of the UGSs in each landscape feature to the supply of public perceptions of each CES, a random forest analysis was performed.

Data source selection

We selected Little Red Book (LRB) as the primary data source for analyzing youth perceptions of urban green spaces in central Beijing. LRB, launched in 2013, is a leading Chinese social networking platform with over 300 million registered users and 85 million monthly active users (Pu 2018). Its user-generated content—comprising captions, images, and metadata-enabled notes—provides a rich dataset for studying real-life environmental experiences. Two main reasons justify our choice of LRB over other leading social media platforms:

Demographic Alignment with Target Population: LRB's user base demonstrates strong alignment with our target population—83.31% of its 85 million monthly active users are aged 18–34 (Pu 2018), which significantly higher than other popular Chinese social media platforms, such as Weibo(52.1%) and Wechat (41.6%) (Appendix 1) (iiMedia Report 2023). Instagram's exclusion is further justified by its regulatory inaccessibility in mainland China (Pu 2018), limiting its relevance for domestic research.

Content suitability: LRB users predominantly share authentic lifestyle experiences, which makes 78% of LRB posts are categorized as "lifestyle" content featuring users' detailed descriptions of real-life experiences, unlike Weibo's focus on public discourse (e.g., news, celebrity gossip) or WeChat's private social networking (Appendix 1) (LRB White Paper2023, Chen et al. 2022). Besides, 89% of LRB notes include GPS metadata (Appendix 1), enabling granular spatial analysis of user-environment interactions—critical for mapping perceptions of green spaces. In contrast, geotagging on Weibo and WeChat is opt-in and sparse (CNNIC 2022). Thus, LRB could provide rich data sources for users' personal experiences for UGSs and relevant spatial information.

Therefore, we selected LRB as the primary social media data source as it was aligned with our target population and research objectives.

Data collection

We firstly used LRB's API to get original note data in this investigation. Many active users provide photos with geographical information in public notes to create a CES-aware city database.

We collected data from Little Red Book's notes from May 2020 to December 2021, when Beijing's public outdoor activities were open after the pandemic, indicating public yearning for nature post-covid. In a preliminary search utilizing CES-related keywords and "Beijing" as keywords, we found many prospective CES service spots in Beijing. These keywords include "outdoor, outdoor recreation, outdoor experiences, excursion, forest, park, water, wetland, river, lake, and grass" (in Chinese). Based on the tags, we found 9022 geotagged notes made by LRB users to share their thoughts and views on the topics. Each note includes photos, texts, themes, the point of interest (POI), and latitude and longitude.

Next, we excluded geotagged LRB notes outside central Beijing via following steps: (1) Boundary Definition: As mentioned in the case study areas, central Beijing includes the main urban areas of six districts (e.g., Dongcheng, Xicheng). We determined the administrative boundaries of these districts, which define the central Beijing boundary, based on Beijing's official land use vector map (2021) from local Municipal Planning Administrations. (2) GIS Screening: we imported the 9,022 geotagged LRB notes into ArcGIS Pro 3.0 and cross-referenced them with the above boundary of central Beijing. Notes outside boundary were automatically flagged for removal (n = 423).(3) Manual screening: we invited two trained researchers independently reviewed the remaining LRB notes. They removed those with geotags within the case study area but content indicating locations outside central Beijing (e.g., notes with central Beijing geotags but contents about "Great Wall spring outing").

The final dataset contains 8,580 rigorously validated notes about central Beijing (95.2% retention rate). The dataset includes 62,138 CES nature perception photographs taken between September 2020 and September 2021, marking the first year that LRB users UGSs experiences in central Beijing. We used validated photos to create a Beijing CES perception database. Similar to prior studies, we grouped geotagged note POI as Table 1.

Table 1 Typology of UGSs in various POI in central Beijing relating to the geo-tagged notes of CES

Landscape features	POI
Land cover human made	Building
	Historical buildings
	Built environment (including small green spaces)
	Street
	Historical square
	Parking sites
	Boulevard,
	Sports facilities
	Urban parks (city level park)
Land cover nature	Forests
	Protected habitat and scenic area
	Blue space
	Nonforested greenspace
	Meadow

Table 2 Typology of youth perceived CESs

Category of CES	Description
Artistic and cultural appreciation	Artistic or cultural expressions and appreciation Photographs representing people in artistic activities (e.g. painter, sculptor), cultural activities (e.g., artisanal fisheries, folk dancing) or their products (e.g. painting, pottery)
Historical monuments	Historical monuments photographs depicting historical infrastructure (e.g. historical buildings, ruins)
Landscape appreciation	Landscape appreciation photographs for which the main focus is a wide and large-scale view of the landscape
Nature appreciation on animals, plants	Nature appreciation photographs focusing on animals, plants or other living organisms
Natural structures and monuments	Natural structures and monuments photographs depicting a specific and well-designed landscape structure (e.g. waterfall, cave)
Religious and spiritual values	Religious, spiritual or ceremonial activities and monuments photographs representing religious or spiritual monuments or activities (e.g. church, indigenous ritual)
Research and education	Research and education photographs showing research or education activities (e.g. researcher, school children) or equipment (e.g. binoculars, traps, notebook, etc.)
Social recreation	Social Recreation Photographs that represent groups of people in an informal or non-dedicated recreative (i.e. not sport) social environment
Sport recreation	Sports recreation Photographs presenting one or more individuals in a dedicated sport activity
Other photographs	Other Photographs that do not fit any of the above criteria

Deep learning models for recognizing photos' content

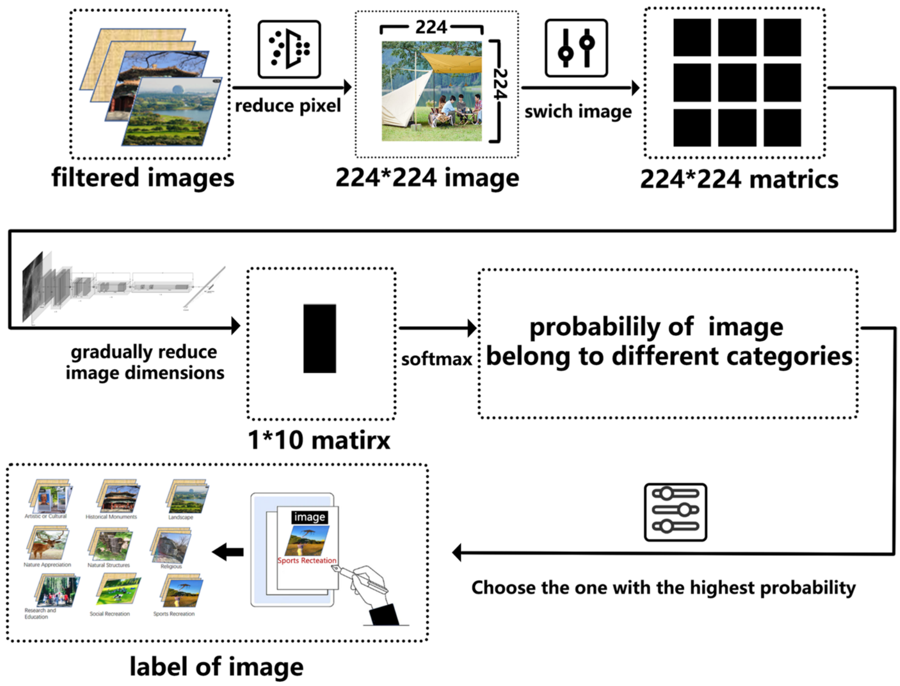
Classification of cultural ecosystem services

Based on the characteristics of high-density megacities and areas with abundant cultural and natural resources, the CES categories in our study were initially adapted from the Millennium Ecosystem Assessment (MEA 2005) and Richards and Friess' framework in Singapore City (2015) and partially inspired by Retka's framework of CES classification (2019) in conservation areas. Sports recreation services were divided from social recreation to represent youth fitness interests. To express local cultural and natural diversity, we classified aesthetic values into artistic and cultural appreciation, landscape appreciation, nature appreciation on animals and plants, natural structures, and monuments according to Retka's CES classification (2019). After identifying CES types, each shot was categorized into 10 groups (Table 2). The cautious approach was to assign each shot to a single CES category. If a photo could represent multiple CES, the rule of thirds was used to select the most prominent CES within the image frame (Rowse 2012). This is the most common method for assessing CES in photographs. In short, the most noteworthy CES dominated the image.

Image content analysis

For image recognition and classification in this study, we used the Resnet model (Zhang et al. 2019). Resnet is a residual network structure based on convolutional neural networks (He, Zhang et al. 2016), which can effectively overcome the issues of learning efficiency reduction and echelon disappearance due to deeper network depth, thereby allowing the model to have deeper layers. The residual network structure is simple to optimize and highly accurate. The most common Resnet models are the Resnet34, Resnet50, and Resnet101 (Nagpal et al. 2022). Among the ResNet variants, ResNet50 was selected for its optimal balance between model depth (50 layers) and parameter efficiency. We have opted for the Resnet50 model structure because it achieves higher precision with fewer training resources. In the Resnet50 model, the input image is resized, the image's dimension is gradually reduced by matrix transformation, a 1×10 matrix is output, and the matrix's values can be interpreted as the probability of the image under this classification after a softmax. As the label of

Fig. 2 Image content analysis using the ResNet50 model for classifying CES categories



this image generated by the model, we select the option with the highest probability (Fig. 2).

For the training process, we adopted a two-stage training strategy combining self-supervised learning with supervised fine-tuning to achieve high classification accuracy while addressing data scarcity in our domain. We used the pretrained ResNet50 based on the SimCLRv2 framework to extract generic visual features through contrastive learning with a temperature coefficient $\tau = 0.5$ and random data augmentations (horizontal flipping/color jittering). Since the SimCLRv2 framework typically pretrains ResNet50 on the ImageNet dataset, which contains over 1 million images, this endows the model with strong generalization ability to distinguish image's content and resistance to data bias (Chen, Ting et al.2020). Pervious studies indicated this method could improves the accuracy of Resnet50 models to more than 80% with only 1% of labeled data, surpassing fully supervised baselines (76.1%) while maintaining model robustness. This demonstrates the framework's effectiveness in semi-supervised learning scenarios with a small amount of labelled data (Chen et al. 2020), which is superior to the standard supervised training methods that require the entire labeled dataset. Then we labeled around 2000 images for fine-tuning and further validation. For the target CES classification task, we fine-tuned the model using 1,864 human-labeled images as task-specific training set (3% of the total 62,128 collected samples). The fine-tuning process furthermore enhanced the model's performance on our CES classification task.

Table 3 Typology and numbers of training set for classifying CES categories

Typology of the CES	Numbers	Proportion of all images (%)
Artistic/cultural appreciation	163	9.5
Historical monuments	139	8.15
Landscape appreciation	148	8.67
Nature appreciation on animals, plants	207	12.02
Natural structures and monuments	85	4.87
Religious, spiritual monuments	118	6.81
Research and education	66	3.81
Social recreation	249	14.59
Sport recreation	147	8.55
Other photographs	394	23.04

For the validation process, as previous studies suggested that normally (1/)×100% of examples should be used for cross validation testing (Amari et al. 1997), we therefore randomly selected 173 images from the labeled image dataset to form a stratified validation set

(Table 3), ensuring no overlap with the training data. We employed fivefold stratified cross-validation during training and consistently applied data augmentations during validation. The model demonstrated strong performance on the validation set, achieving 85.3% accuracy, 0.827 macro F1 score, and 0.864 weighted F1 score (Amari et al. 1997). These results indicate that our model not only balances parameter efficiency with classification accuracy but also effectively controls for potential biases through data partitioning and domain adaptation strategies.

Despite the effectiveness of the ResNet50 model with SimCLRv2, potential biases can affect classification accuracy. Dataset bias is a primary concern, as model performance heavily depends on the diversity of training images. Limited and imbalanced labeled data may lead to biased predictions, affecting generalization to underrepresented categories. To mitigate these biases, we used pretrained model based on various plain image dataset which can effectively avoid data bias and enhance models generalisation ability.

Kernel density analysis and spatial statistical analysis in blocks

To create heatmaps, which are effective for visually representing the spatial clustering of the mapped ecosystem values, we selected Kernel density analysis. Kernel density is a point pattern analysis techniques available in ArcGIS that estimates the density of point events in a given area, and this method describes the spatial extent and intensity of CES distribution by delivering a smooth density surface based on the point density ultimately outputted as a raster dataset. The density value for each pixel represents the sum of the values of all kernel surfaces superimposed on the center of the raster image element. Here, we set the output image element size to 100 m and the search radius to 500 m. In social science, Kernel density has been considered as established and robust method for hotspot mapping (Alessa et al. 2007). Moreover, in CES studies, it also has been taken as a common practice applied (Xu et al. 2020; Garcia-Martin et al. 2017; Fagerholm et al. 2016; Zhang et al. 2022). According to Brown (2008), as opposed to inverse distance weighted interpolation method which often producing "spurious and no-intuitive" pattern did not align with what respondents actually mapped, kernel density makes no assumptions about unsurveyed locations, simply smoothing the density of the collected points, which results in more reliable representation of participate input.

To understand the CES spatial distribution for each street block in the high-density urban context, we summed the total number of the nine classified CES points based on the boundary of street blocks' division of central Beijing.

Random forest analysis

We employed a random forest (RF) model to reveal the complex, nonlinear mechanisms through which urban green spaces (UGSs) influence CES hotspots, as mediated by their landscape contexts (proxied by points of interest, POIs). The RF framework was selected to address two fundamental challenges in CES-UGS relationships: (1) context-dependent nonlinearity, where UGS contributions exhibit threshold effects (e.g., minimum park size required for gain CESs) and synergistic interactions with surrounding environment features (e.g., green spaces coupled with cultural POIs like historical sites) (Plieninger et al. 2013; Chen et al. 2020), and (2) implicit spatial interdependencies among POIs, where UGS functionality often hinges on co-located amenities (e.g., streets enhancing park accessibility, sports facilities amplifying social visitation) that linear models fail to disentangle (Simon et al. 2023). RF was prioritized over other nonlinear methods due to its interpretable feature importance metrics and robustness when fitting with limited spatial data (Hastie et al. 2009).

POIs were categorized to encapsulate UGS attributes critical for CES provision, including functional traits (e.g., subclassifying green spaces by size and facilities such as protected habitants for supporting biodiversity education) and embeddedness of their contextual environment (e.g., adjacency to historical buildings for cultural heritage and sense of place). We used Mean Decrease Impurity (MDI) to quantify feature importance, revealing how specific UGS configurations drive CES provision across the urban landscape. For instance, the high MDI of "parks integrated with historical buildings" highlights their role as cultural anchors, where interpretive infrastructure unlocks heritage value, while the low MDI of "isolated green patches" underscores their limited CES impact without synergistic amenities like recreational

Table 4 Descriptive statistics of youth perceived CES perceived in nine categories

CES	Numbers Of photos	Proportion of photos for each kind of CES categories (%)	Numbers of users	Proportion of users for each kind of CES categories (%)
Artistic/cultural appreciation	4350	7	722	8
Historical monuments	4350	7	722	8
Landscape appreciation	5592	9	902	10
Nature appreciation on animals, plants	6835	11	902	10
Natural structures and monuments	2486	4	541	6
Religious, spiritual monuments	3107	5	632	7
Research and education	1864	3	361	4
Social recreation	15,534	25	1714	19
Sport recreation	4350	7	902	10
Other photographs	13,670	22	1624	18

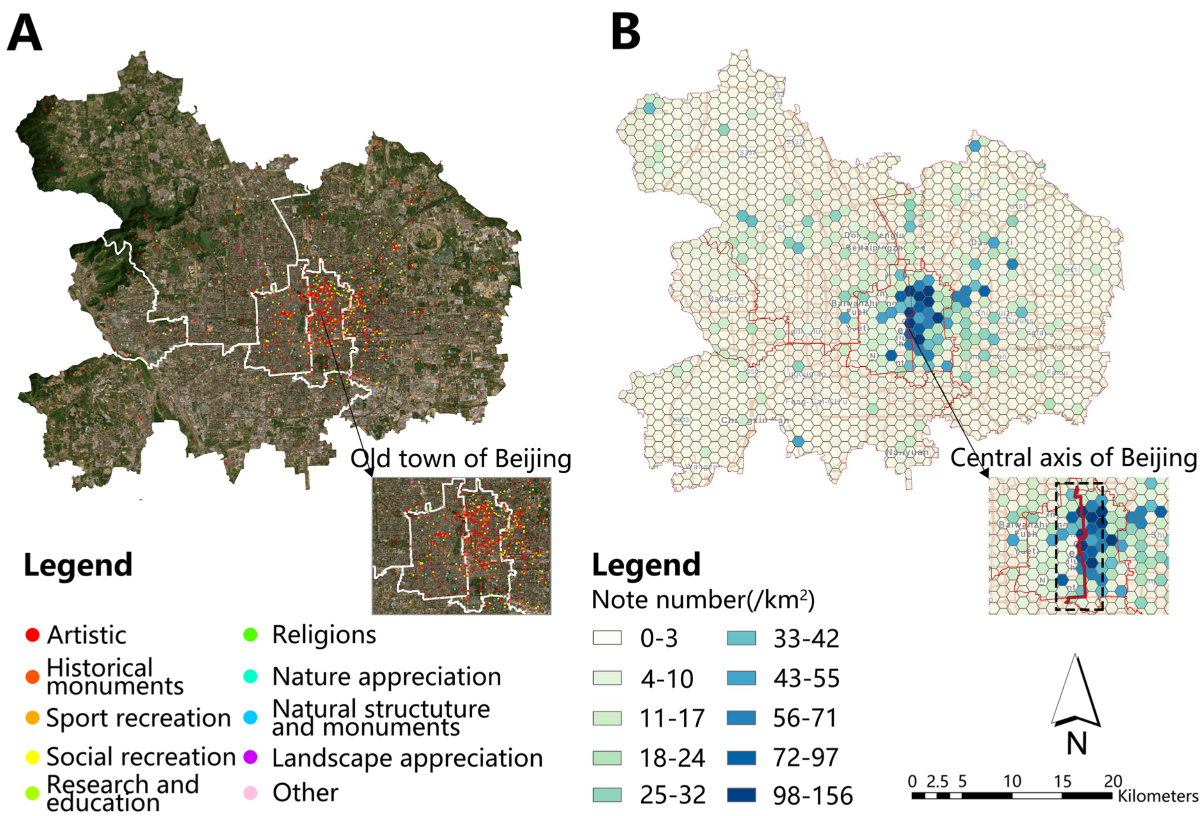


Fig. 3 **A** The point map of geotagged photos classified in nine categories, **B** the spatial density of all geotagged photosas hexagonal binning map

facilities, seating(Menze et al. 2009). We validated these patterns through fivefold cross-validation, confirming stable MDI rankings across iterations (Breiman 2001). The outcome of random forests could reveal how different UGSs featured with multiple POIs disproportionately contribute various CESs.

Results

Descriptive analysis

We finally obtained a distribution of 62,138 multiple types of CES points (Table 4; Fig. 3). In terms of number share, social recreation is by far the most frequently mapped CES point, followed by nature appreciation of animals and plants (11%, n = 6835) and landscape appreciation (9%, n = 5592). The majority of the 9022 users who shared images were engaged in social recreation (19%, n = 1714), followed by landscape appreciation, nature appreciation, and sport recreation (10%, n = 902). Research and education (3%, n = 1864) received the least amount of attention and had the fewest number of maps. In terms of spatial distribution, the study reflected the type distribution of youth perceptions of nine CES in various areas of Beijing's central city (Fig. 3). Significant geographical differences were observed, as the overall number of CES points showed a decreasing trend from the center to the edge of the study area. The capital core area has the highest concentration of CES distribution, and as one moves away from the city center, the number of CES distribution points decreases. The Dongcheng District, as the east part of the capital core area, has the highest concentration of mapped CES points. The Xicheng District, which is the west part of the capital core area, has a more restricted distribution and is concentrated in the area surrounding Beijing's central axis. Other regions had a more dispersed distribution of CES points and a lower total. The two districts with the lowest number of CES points were Shijingshan District and Fengtai District.

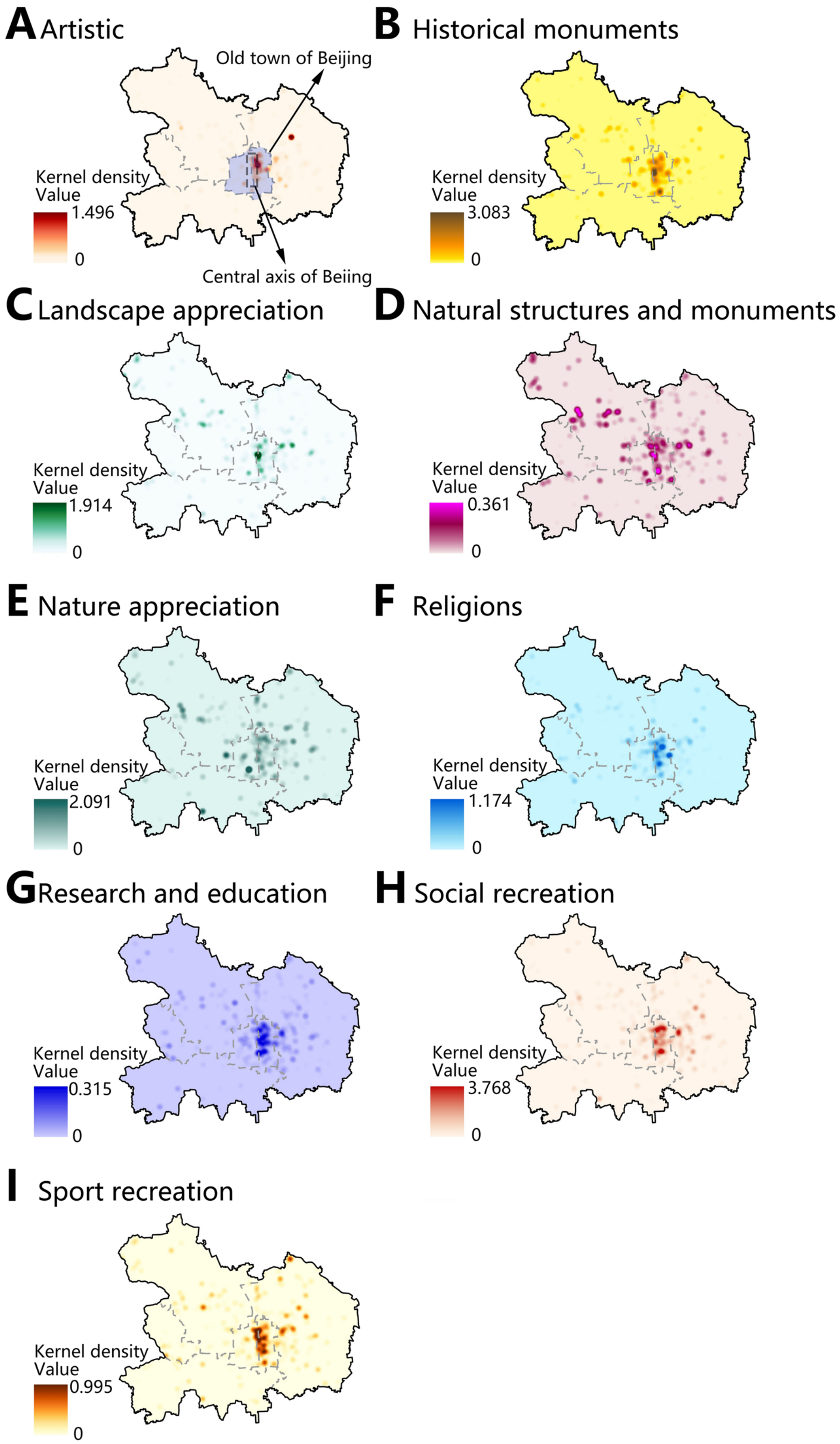
Spatial distribution of various youth mapped CES

In our study area, the distribution of CESs was not uniform, as revealed by our heat map depicting the spatial distribution of all identified CES points, with darker hues denoting a higher intensity of spatial distribution (Fig. 4A–I). Notably, within the old town area of Beijing, there was a discernible concentration of high-density points representing various types of CESs, particularly evident in the eastern sector. In contrast, natural-related CESs, such as landscape appreciation, and nature appreciation on animal and plants perceptions, exhibited a more scattered pattern across central Beijing (Fig. 4E, H, I).

High-density clusters of social recreation were primarily mapped in the old town and eastern-central Beijing, correlated with accessible public spaces with active social recreational environments offering outdoor facilities (e.g., riverside plazas, multifunctional parks). Moreover, the highest densities of natural appreciation and natural structures were also observed, featuring with the traditional mountain landscapes, greenways, and royal gardens, dominating the northwest of central Beijing and iconic lake in old town (Fig. 4D, E). Artistic value points formed pronounced clusters in the eastern part of Beijing (Fig. 4A), driven by cultural hubs integrating museums, galleries, and community-led art revival initiatives. Similarly, sports recreation hotspots concentrated in both the western and eastern urban sectors of central Beijing, closely associated with legacy Olympic venues, Asia Games village, greenway systems equipped with comprehensive sports facilities. Historical value exhibited the strongest spatial aggregation, with hotspots densely aligned along Beijing's central axis—the core of the old town where imperial-era heritage and architectural landmarks dominate (Fig. 4B). Additionally, the highest density of religious and spiritual significance points localized within the old town area, often proximate to temples, churches, and memorial parks featuring curated green spaces (Fig. 4F).

We further revealed the spatial patterns of CES benefits across 145 blocks (Fig. 5A-I) to unravel the micro-scale clustering of CESs.Artistic value (Fig. 5A) formed dual clusters in Chaoyang District's Jiuxianqiao and Maizidian blocks (77–141 points), driven by adaptive reuse of industrial spaces into galleries and community-led art initiatives. Historical value (Fig. 5B) exhibited extreme aggregation along Beijing's central axis, where blocks such as Shichahai, Donghuamen, and Jingshan recorded 130–270 points per block, encompassing 92% of the city's nationally protected heritage sites, with Hutong historic neighborhoods and imperial-era architecture coexisting with revitalized historic neighborhoods. Nature-related CESs, including landscape appreciation and natural structures (Fig. 5C–E), displayed a bipolar distribution: northwest mountain blocks (Xiangshan) peaked seasonally with 68–127 points linked to autumn foliage, while engineered green spaces in eastern blocks (Olympic Park) achieved 67–114 points through designed wetlands, surpassing 76%

Fig. 4 Kernel density heat maps of the spatial distribution of: **A–I** each individual CES. A high density of points is visualized in dark brown and a low density in light orange. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



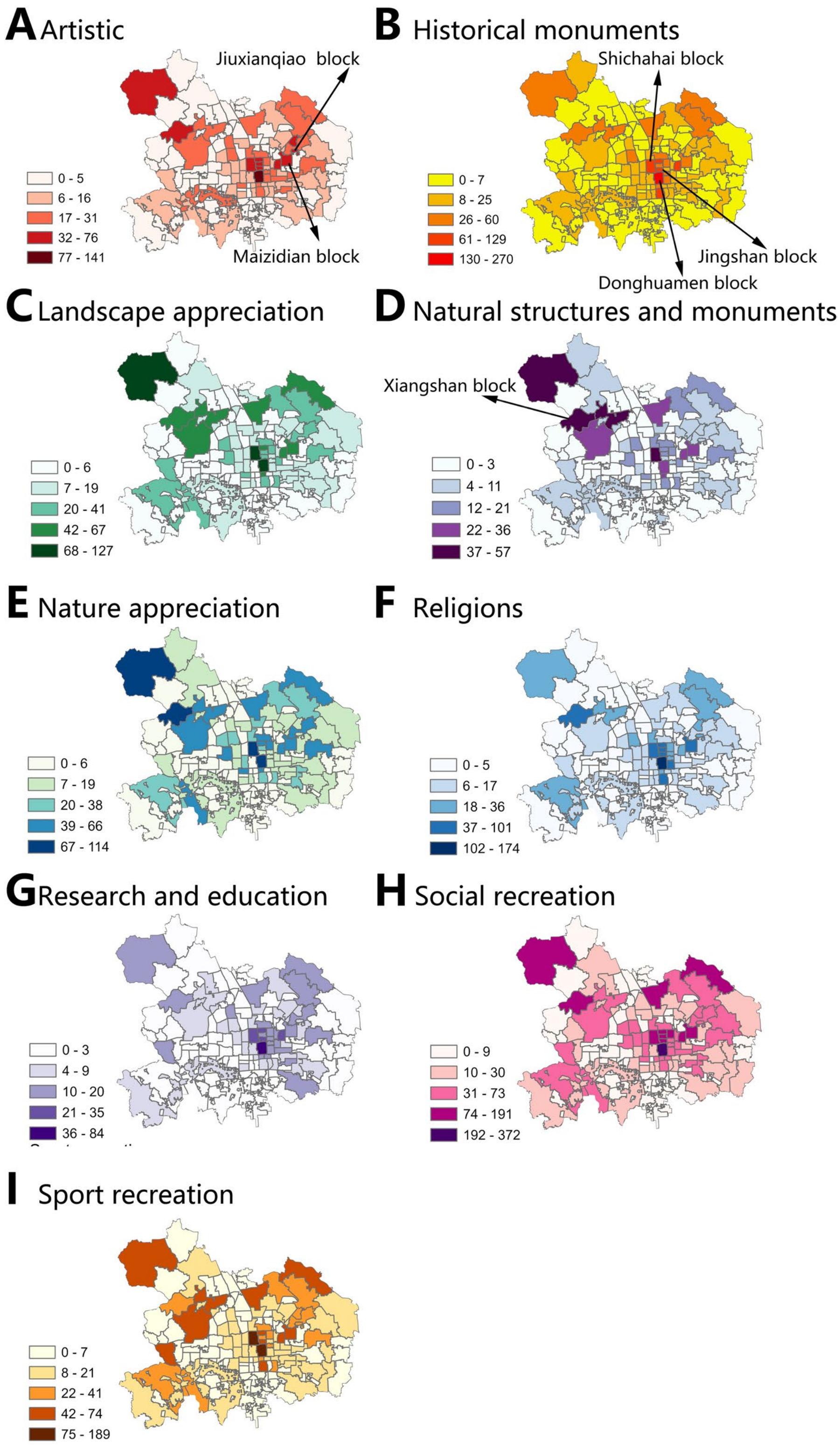


Fig. 5 Spatial distribution of each CES in different blocks in central Beijing. **A–I** Each individual CES

of Xiangshan's density despite being 58% smaller area. Religious significance (Fig. 5F) concentrated in temple-dense blocks near Yonghe Lama Temple (Dongcheng District, 102–174 points), contrasting with memorial parks (37–101 points) that emphasize commemorative rituals. Social and sports recreation (Fig. 5H, I) clustered in hybrid-use zones: traditional commercial blocks (Donghuamen, 192–372 points) in Dongcheng District and legacy event blocks (Yayuancun, former Asian Games Village) in Chaoyang Distrct. To sum up, this pronounced spatial segregation between cultural and natural CES-providing blocks underscores Beijing's dual-track urbanization—heritage preservation in the core perpetuates cultural service clusters, while topographic constraints and compensatory green infrastructure push nature-related services to engineered peripheries.

The contribution of various landscape features for hotspots of various kinds of CES

The variable importance plots for all CES and for each CES individually derived from the random forest model were illustrated in Fig. 6A–I. All plots illustrate the high importance of UGSs in the POIs of human-made landscape features relative to natural features. Examining the plot for all CES, the largest disparity in POI contribution occurred between buildings, urban parks, and other POIs.

Urban parks played the most important role in nature-related CESs, such as landscape appreciation (MDI importance = 0.273), nature appreciation of animals and plants (MDI importance = 0.306), and natural structures and monuments (MDI importance = 0.428) (Fig. 6C–E). Additionally, urban forests (MDI importance = 0.114 for natural structures and monuments) and blue spaces (MDI importance = 0.078 for natural structures and monuments) contributed significantly to these nature-related CESs. In contrast, the contributions of UGSs in natural features (e.g., blue spaces, forests) to cultural-social related CESs (e.g., social recreation, artistic value) were substantially lower than those of UGSs in human-made features (e.g., buildings) (Fig. 6A, B, F–I).

For cultural- and social-related CES, UGSs in buildings (e.g., roof gardens, vertical greening, and micro green spaces) were the dominant contributors. These UGSs exhibited the highest importance for artistic value (MDI importance = 0.250), followed by urban parks (MDI importance = 0.238) and UGSs in built environments (e.g., pocket parks and community green spaces; MDI importance = 0.160). Similarly, UGSs in buildings contributed more to religious values (MDI importance = 0.329) than urban parks (MDI importance = 0.131). Notably, UGSs in buildings also demonstrated the highest importance for social recreation (MDI importance = 0.561), while urban parks (MDI importance = 0.093), built environments (MDI importance = 0.088), and streets (e.g., boulevards; MDI importance = 0.064) showed progressively lower contributions.

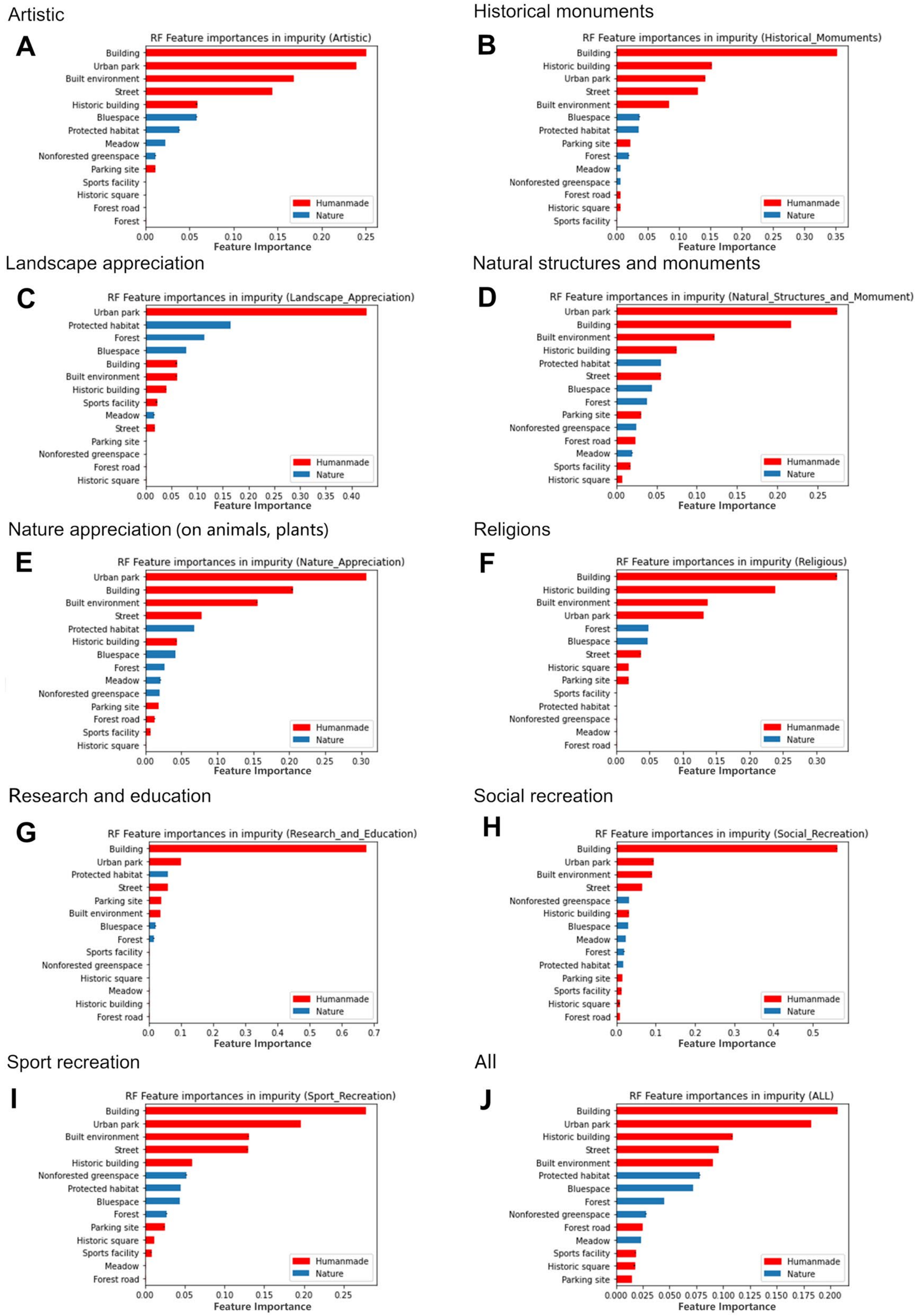
Discussion

Youth-perceived CES in central Beijing

Our study explored the spatial distribution of youth-perceived CESs in central Beijing using crowdsourced social media imagery, analyzed with deep learning techniques. The results revealed an uneven geographic distribution, with nature and landscape appreciation clustering in the northwest, eastern, and old-town areas, while social recreation, artistic expression, and sports activities concentrated along the central axis of the old town. These findings offer actionable insights for urban green space (UGS) planning and management that address youth interests.

Unlike traditional CES studies that rely on expert evaluations or PPGIS-based methods (e.g., Thiagarajah et al. 2015; Xu et al. 2020), which capture public perceptions based on predefined categories and expert knowledge, our crowdsourced imagery approach provides a dynamic, real-time view of youth interactions with UGSs. These traditional methods tend to be more static, offering limited geographical coverage and a more controlled sampling process. In contrast, our approach leverages spontaneous, user-generated data, allowing us to capture a broader, more diverse sample and providing new insights into how youth engage with urban green spaces.

While our findings align with previous research emphasizing the importance of recreational and aesthetic values (Thiagarajah et al. 2015; Xu et al. 2020), we observed that historical and educational values were lower in our study compared to earlier research. This difference highlights how youth in


Artistic
A
RF Feature importances in impurity (Artistic)
Building
Urban park
Built environment
Street
Historic building
Bluespace
Protected habitat
Meadow
Nonforested greenspace
Parking site
Sports facility
Historic square
Forest road
Forest
Humanmade
Nature
0.00 0.05 0.10 0.15 0.20 0.25
Feature Importance
Historical monuments
B
RF Feature importances in impurity (Historical_Momuments)
Building
Historic building
Urban park
Street
Built environment
Bluespace
Protected habitat
Parking site
Forest
Meadow
Nonforested greenspace
Forest road
Historic square
Sports facility
Humanmade
Nature
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35
Feature Importance
Landscape appreciation
C
RF Feature importances in impurity (Landscape_Appreciation)
Urban park
Protected habitat
Forest
Bluespace
Building
Built environment
Historic building
Sports facility
Meadow
Street
Parking site
Nonforested greenspace
Forest road
Historic square
Humanmade
Nature
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40
Feature Importance
Natural structures and monuments
D
RF Feature importances in impurity (Natural_Structures_and_Momument)
Urban park
Building
Built environment
Historic building
Protected habitat
Street
Bluespace
Forest
Parking site
Nonforested greenspace
Forest road
Meadow
Sports facility
Historic square
Humanmade
Nature
0.00 0.05 0.10 0.15 0.20 0.25
Feature Importance
Nature appreciation (on animals, plants)
E
RF Feature importances in impurity (Nature_Appreciation)
Urban park
Building
Built environment
Street
Protected habitat
Historic building
Bluespace
Forest
Meadow
Nonforested greenspace
Parking site
Forest road
Sports facility
Historic square
Humanmade
Nature
0.00 0.05 0.10 0.15 0.20 0.25 0.30
Feature Importance
Religions
F
RF Feature importances in impurity (Religious)
Building
Historic building
Built environment
Urban park
Forest
Bluespace
Street
Historic square
Parking site
Sports facility
Protected habitat
Nonforested greenspace
Meadow
Forest road
Humanmade
Nature
0.00 0.05 0.10 0.15 0.20 0.25 0.30
Feature Importance
Research and education
G
RF Feature importances in impurity (Research_and_Education)
Building
Urban park
Protected habitat
Street
Parking site
Built environment
Bluespace
Forest
Sports facility
Nonforested greenspace
Historic square
Meadow
Historic building
Forest road
Humanmade
Nature
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
Feature Importance
Social recreation
H
RF Feature importances in impurity (Social_Recreation)
Building
Urban park
Built environment
Street
Nonforested greenspace
Historic building
Bluespace
Meadow
Forest
Protected habitat
Parking site
Sports facility
Historic square
Forest road
Humanmade
Nature
0.0 0.1 0.2 0.3 0.4 0.5
Feature Importance
Sport recreation
I
RF Feature importances in impurity (Sport_Recreation)
Building
Urban park
Built environment
Street
Historic building
Nonforested greenspace
Protected habitat
Bluespace
Forest
Parking site
Historic square
Sports facility
Meadow
Forest road
Humanmade
Nature
0.00 0.05 0.10 0.15 0.20 0.25
Feature Importance
All
J
RF Feature importances in impurity (ALL)
Building
Urban park
Historic building
Street
Built environment
Protected habitat
Bluespace
Forest
Nonforested greenspace
Forest road
Meadow
Sports facility
Historic square
Parking site
Humanmade
Nature
0.000 0.025 0.050 0.075 0.100 0.125 0.150 0.175 0.200
Feature Importance

◂**Fig. 6** **A–I** Variable importance plots for all CES. Plots were derived from random forest models (%IncMSE is the increase in mean of the error of a tree)

high-density urban environments may prioritize different aspects of CES compared to broader public perceptions, which often place more emphasis on cultural heritage and natural education. The methodological differences—crowdsourced data vs. expert evaluations and PPGIS—likely contribute to this variation, as traditional methods may not capture the spontaneous and socially oriented experiences of youth in urban green spaces.

Additionally, our findings suggest that social recreational value is also an important CES for the younger generation, which aligns with the findings of Seeland et al.(2008), who emphasized the essential contribution of urban green spaces to youth social inclusion. Our findings further align with those of Luo et al. (2023), who found that youth perceive aesthetic values more strongly in forest areas, emphasizing the importance of considering youth preferences when planning urban green spaces. However, we also found that youth-perceived CES related to research and education was notably low (4%) in central Beijing. This contrasts with previous studies where educational values received more recognition (e.g., Thiagarajah et al. 2015). This result is consistent with findings in European cities (Zwierzchowska et al. 2018), where residentsalso highlighted a limited recognition of educational potential in UGSs.

The crowdsourced approach offers a more immediate perspective of youth interactions with UGSs, possibly revealing a gap in natural education that is not as evident in more traditional assessments. To address this gap, we recommend enhancing natural educational offerings, such as biodiversity education programs and facilities, for young people within central Beijing's UGSs.

The contribution made by UGSs in various landscape features for the supply of each CES

Our findings highlight that urban green spaces (UGSs) integrated into human-made landscapes are particularly important for providing various youth-perceived CESs in central Beijing, aligning with previous research on CES studies based on PPGIS (Plieninger et al. 2013; Ridding et al. 2018).

Specifically, green spaces within built-up areas, historical buildings, and streets significantly contribute to multiple youth-perceived CES benefits, such as social and sports recreation, cultural heritage, religious experiences, and natural appreciation of plants and animals. This is consistent with previous studies that emphasize the role of micro-green spaces and pocket parks in providing diverse CES benefits (Xu et al. 2020; Yang and Hong 2023). However, while we observe the importance of UGSs in built environments for youth, Zhang et al. (2022), who used VEP photography methods with young participants, found that landscape sketches and waterbodies in urban parks are the most important landscape features for providing CES to young people, which contrasts with our findings. Such discrepancy can be attributed to the methodological differences: while VEP photography studies assess youth perceptions during leisure time (e.g., CES photo surveys in parks), our crowdsourced social media imagery approach captures spontaneous, real-time perceptions of youth during their daily lives. As young people in high-density cities typically have limited opportunities to access green spaces due to long work hours, the UGSs integrated into built environments that they encounter during work or commute likely play a more significant role in their CES perception (Liu et al. 2023a, b).

Besides, our results highlight that urban parks in central Beijing significantly contribute to landscape appreciation, nature appreciation, and the value associated with natural structures and monuments. This can be attributed to their notable landscape design and management, such as the renowned red leaf scenery and mountainous landscape at Xiangshan scenic area, spring cherry blossom views, and rich bird habitats (Xu et al. 2023). Additionally, these parks usually play essential roles in providing aesthetic value and promoting sports recreation. This finding aligns with earlier research based on suvery and social media texts analysis emphasizing urban parks as crucial venues for aesthetic experiences and sporting activities (Cheng et al. 2023). Peters et al. (2017) also highlighted urban parks as vital community spaces for outdoor leisure and educational activities for children. Nevertheless, compared to UGSs within building areas, our results pointed out that urban parks in central Beijing are less critical for social recreation and educational purposes. This discrepancy may result from insufficient socialization events and facilities in these parks (Xu et al. 2023).

Notably, our study found that bluespaces in Beijing provided significantly fewer recreational benefits compared to other types of urban green spaces (UGSs). This result contrasts with previous CES studies based on PPGIS which emphasized bluespaces as essential landscape features for delivering multiple CES, especially recreational values (Gerstenberg et al. 2020; Xu et al. 2020). This discrepancy may seem unexpected given earlier studies highlighting the recreational value of urban water bodies, including activities such as fishing, camping, and swimming (Pearson et al. 2017; Grzyb et al. 2023; Vierikko and Yli-Pelkonen 2019). However, bluespaces in Beijing have not served as significant sources of recreational value, due to numerous administrative restrictions aimed at protecting the environmental quality of local water bodies. Since 2020, the Beijing Water Bureau (BWB 2020) has enforced restrictions against activities like swimming, fishing, large gatherings, barbecues, and picnics near public water bodies, including rivers and lakes. In accordance with this, we also observed substantial complains from the notes, revealing real-time public perceptions, complains and experiences on social media, which may not be as evident in traditional methodologies like PPGIS or VEP photography.

Implications for UGSs management in high-density urban context

Our study provides valuable insights into the CES perceived by youth in central Beijing, filling the gap in understanding the benefits young populations derive from urban green spaces (UGSs). By leveraging social media imagery and deep learning techniques, we captured a comprehensive dataset that offers a nuanced view of youth perspectives. This crowdsourced approach, combined with spatially visualized CES data, helps urban planners identify areas with high potential for cultural, aesthetic, and recreational investments, ultimately enhancing UGS quality. Based on our findings, we present the following specific policy implications and recommendations to improve UGSs and maximize their benefits for youth in central Beijing:

(1) Prioritize micro-green spaces in buildings and built environments: urban planners and managers should prioritize investments in micro-green spaces in buildings and built environments(e.g., roof gardens, vertical greening, pockets park) to enhance young people's exposure to green spaces and increase their social engagement, thereby helping to alleviate their anxiety and depression in modern urban lives (Reece et al. 2021). Such efforts can be implemented through regulatory innovations and incentives in future UGSs management, including mandating rooftop gardens in building codes of central Beijing or offering tax rebates for developers adding pocket parks in communities, and vertical greening projects that include art installations and social seating-like Copenhagen's and Aarhus' green roof initiatives (Green Roof Policy—Copenhagen 2023).

(2) Enhance social functionality in urban parks: redesign underutilized park into multi-functional spaces to enhance their contributions to social recreation, artistic expression, and community engagement by incorporating facilities spaces and events for artistic activities (e.g., outdoor art festival), social gatherings (e.g., theme markets, picnic zones), educational programs and natural appreciation workshops. Encouraging natural appreciation, sports recreation, artistic activities, and public events in these parks could effectively improve community engagement and social interaction of young people (Zheng et al. 2023).

(3) Revitalize bluespaces through adaptive governance: urban planners can implement zoned management by designating vibrant zones for hotspots areas like Houhai lake, offering timed access through management permits to enhance sports and social recreational opportunities. Buffer zones with features like bird-watching platforms can be created in areas like urban rivers to provide natural education value, while core areas are strictly protected, balancing conservation and recreation to make blue spaces more accessible to youth while preserving environmental quality.

(4) Optimize UGS quality for guiding visitor management: urban planner could ultimately improve site-specific UGSs quality based on public perception. Lower perceived intensity areas could be prioritized for ecological functions like bio-

diversity conservation or storm water regulation. In addition, identified CES hotspots could guide visitor management strategies by alleviating pressure on highly visited areas and promoting exploration of less-visited sites through enhanced local recreation opportunities and events (Sinclair et al. 2020).

(5) Leverage digital platforms for participatory planning: create a youth green co-design platform to allow young people to contribute to CES mapping, vote on UGS upgrades, and directly influence park improvements based on their needs. This could also be combined with current social media platforms. For example, organizing an annual "Most Beautiful Roof Garden/Pocket Park/Urban Park" contest through the LRB platform could enhance public participation in green space management.

Limitations and further research

Our study contributes to the management of UGSs in high-density urban areas by examining the CES demands of young people using social media photos and deep learning methods. Nevertheless, it is important to acknowledge the limitations inherent in social media data. Subjective judgment inevitably occurs during keyword development, image selection, and the association of images with specific CES categories. For example, a photo might depict both aesthetic elements and recreational activities, such as kayaking, but the analysis might primarily highlight aesthetic aspects, potentially overlooking the recreational aspects. Similar subjective judgments are noted in existing CES studies utilizing image context analysis (Cao et al. 2022). To address this issue, future research should integrate manual image analysis more comprehensively with automated machine learning techniques, thus achieving a more nuanced understanding of youth-perceived CESs.

Conclusion

This study utilized social media imagery and deep learning techniques to assess the spatial distribution and youth perceptions of CESs in central Beijing, offering critical insights for urban green spaces (UGSs) management in high-density urban contexts. Our results revealed that social recreation is the most frequently perceived CES, followed by nature appreciation on animals and plants. The spatial analysis highlighted that social recreation is concentrated in the old town and eastern areas, whereas nature appreciation is more widely distributed throughout central Beijing. Our findings underscore the significant role of UGSs integrated into built environments, such as micro-green spaces and residential greenness, in providing CESs related to social recreation, historical value, and artistic expression. In contrast, urban parks and other nature-focused landscape features were more influential in promoting landscape and nature appreciation, suggesting that UGSs need further enhancements to better support social recreational services. Interestingly, bluespaces in Beijing provided fewer recreational benefits than expected, suggesting the need for better management strategies that balance environmental protection and recreational use. The crowdsourced imagery approach offers a dynamic view of youth engagement, contrasting with traditional methods and providing a more comprehensive understanding of youth preferences. To improve the quality and accessibility of UGSs, urban planners and policymakers should focus on specific actions: (1) Invest in micro-green spaces like rooftop gardens and pocket parks to enhance youth access and social engagement. (2) Enhance social functionality in urban parks: design parks to better support social recreation and artistic activities for young people. (3) Revitalize bluespaces through adaptive governance: implement zoned management to allow regulated access to bluespaces while maintaining environmental quality. (4) Optimize UGS quality for guiding visitor management. (5) Create digital platforms for participatory planning: leverage digital platforms for participatory planning and improvement of UGSs.

This research emphasizes the importance of aligning UGS design with youth needs to improve well-being in high-density urban areas and underscores the value of combining digital and traditional methods for sustainable urban planning.

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Author contributions Haiyun Xu and Jianxuan Duan: worte the orginial draft. Haiyun Xu: conceptualization, methodology, funding acquisition, resources, Renmu Jie: data curation, methodology, software, visualization; Guohan Zhao: writing—review & editing, supervision, visualization, software, investigation. Zhifeng Liu: writing—review and editing, supervision.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare that there are no conflicts of interests.

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